

6.4.8

$$\text{Corr}(X_1, X_N) = \frac{\text{Cov}(X_1, X_N)}{\text{SD}(X_1) \text{SD}(X_N)} = \frac{-k/N^2}{\sqrt{\frac{k(N-1)}{N^3} \cdot \frac{k(N-1)}{N^3}}} = \frac{-k/N^2}{\frac{k(N-1)}{N^2}} = \boxed{-\frac{1}{N-1}}$$

$$X_i \sim \text{Bin}(k, \frac{1}{N})$$

$$\text{Var}(X_i) = k(\frac{1}{N})(1 - \frac{1}{N}) = \frac{k(N-1)}{N^2}$$

$$\begin{aligned} \text{Cov}(X_1, X_N) &= \text{Cov}\left(\sum_{i=1}^k I_i, \sum_{j=1}^k J_j\right) = \sum_{i=1}^k \sum_{j=1}^k \text{Cov}(I_i, J_j) \\ &= \sum_{i=1}^k \sum_{j=1}^k E(I_i J_j) - E(I_i)E(J_j) \\ &= \sum_{i=1}^k \sum_{j=1}^k 0 - \frac{1}{N}\left(\frac{1}{N}\right) = -k\left(\frac{1}{N^2}\right) \end{aligned}$$

6.4.10

$$\text{Corr}(H_{100}, H_{300})$$

$$X = \text{First 100 tosses} \sim \text{Bin}(100, 1/2)$$

$$Y = \text{Last 200 tosses} \sim \text{Bin}(200, 1/2)$$

$$\begin{aligned} \text{Cov}(H_{100}, H_{300}) &= \text{Cov}(X, X+Y) \\ &= \text{Cov}(X, X) + \text{Cov}(X, Y) \\ &= \text{Var}(X) + \text{Cov}(X, Y) \\ &= 100(1/2)(1/2) + 0 \\ &= 25 \end{aligned}$$

$$\begin{aligned} \text{Corr}(H_{100}, H_{300}) &= \frac{\text{Cov}(H_{100}, H_{300})}{\text{SD}(H_{100}) \text{SD}(H_{300})} \\ &= \frac{25}{\sqrt{25(75)}} = \frac{25}{25\sqrt{3}} = \boxed{\frac{1}{\sqrt{3}}} \end{aligned}$$

$$\begin{aligned} \text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) \\ &= 25 + 200(1/2)(1/2) \\ &= 25 + 50 = 75 \end{aligned}$$

6.4.12

$$a) E(X) = \alpha$$

$$E(Y) = \beta$$

$$b) \text{Var}(X) = \alpha(1-\alpha)$$

$$\text{Var}(Y) = \beta(1-\beta)$$

$$c) \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= 0 - \alpha\beta$$

$$= \boxed{-\alpha\beta}$$

$$d) E(D_n - R_n) = E(D_n) - E(R_n)$$

$$= n\alpha - n\beta$$

$$= \boxed{n(\alpha - \beta)}$$

$$e) \text{Var}(D_n - R_n)$$

$$= \text{Var}(D_n) + \text{Var}(R_n) - 2\text{Cov}(D_n, R_n)$$

$$= (\alpha(1-\alpha) + \beta(1-\beta) - 2(-\alpha\beta))n$$

$$= (\alpha - \alpha^2 + \beta - \beta^2 + 2\alpha\beta)n$$

$$= n((\alpha + \beta) - \alpha^2 - \beta^2 + 2\alpha\beta)$$

$$= \boxed{n((\alpha + \beta) - (\alpha - \beta)^2)}$$